Website Data Analysis

1. What patterns or trends can you observe in website sessions and users over time?
2. Which marketing channel brought the highest number of users to the website, and how can we use this insight to improve traffic from other sources?
3. Which channel has the highest average engagement time, and what does that tell us about user behavior and content effectiveness?
4. How does engagement rate vary across different traffic channels?
5. Which channels are driving more engaged sessions compared to non-engaged ones, and what strategies can improve engagement in underperforming channels?
6. At what hours of the day does each channel drive the most traffic?
7. Is there any correlation between high traffic (sessions) and high engagement rate over time?

In [1]:
```python
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
```

In [15]:
```python
import pandas as pd
from ydata_profiling import ProfileReport
```

In [2]:
```python
df = pd.read_csv(r"C:\Users\sahil\Downloads\data-export (1) (1).csv")
```

In [16]:
```python
df = pd.read_csv(r"C:\Users\sahil\Downloads\data-export (1) (1).csv")

# Create profile report
profile = ProfileReport(df)

# Save report as HTML
profile.to_file("report.html")
```

```
Summarize dataset:   0%|          | 0/5 [00:00<?, ?it/s]
  0%|                                                  | 0/10 [00:00<?, ?it/s]
 10%|████                                      | 1/10 [00:00<00:00,  9.69it/s]
 20%|████████                                  | 2/10 [00:00<00:01,  6.08it/s]
 60%|████████████████████████                  | 6/10 [00:00<00:00, 12.73it/s]
100%|██████████████████████████████████████████| 10/10 [00:00<00:00, 11.86it/s]
Generate report structure:   0%|          | 0/1 [00:00<?, ?it/s]
Render HTML:   0%|          | 0/1 [00:00<?, ?it/s]
Export report to file:   0%|          | 0/1 [00:00<?, ?it/s]
```

In [18]:
```python
import webbrowser
webbrowser.open("report.html")
```

Out[18]:
```
True
```

In [23]:
```python
df.head()
```

Out[23]:

	# --	Unnamed: 1	Unnamed: 2	Unnamed: 3	Unnamed: 4	Unnamed: 5	Unnamed: 6	Unnamed: 7	Unnamed: 8	Unnamed: 9
0	Session primary channel group (Default channel...	Date + hour (YYYYMMDDHH)	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session	Engagement rate	Event count
1	Direct	2024041623	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	0.48	1402
2	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.4943820224719100	1147
3	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.49356223175965700	1069
4	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	0.48828125	1044

In [24]:
```python
df.columns = df.iloc[0]
df = df.drop(index=0).reset_index(drop = True)
df.columns = ["channel group", "DateHour", "Users", "Sessions", "Engaged sessions",     "Average engagement time per session",
       "Engaged sessions per user",      "Events per session     ", "Engagement rate",   "Event count"]
```

In [25]:
```python
df.head()
```

Out[25]:

	channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count
0	Direct	2024041623	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	0.48	1402
1	Organic Social	2024041719	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.4943820224719100	1147
2	Direct	2024041723	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.49356223175965700	1069
3	Organic Social	2024041718	187	256	125	32.16015625	0.6684491978609630	4.078125	0.48828125	1044
4	Organic Social	2024041720	175	221	112	46.918552036199100	0.64	4.529411764705880	0.5067873303167420	1001

In [26]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
 #   Column                               Non-Null Count  Dtype 
---  ------                               --------------  ----- 
 0   channel group                        3182 non-null   object
 1   DateHour                             3182 non-null   object
 2   Users                                3182 non-null   object
 3   Sessions                             3182 non-null   object
 4   Engaged sessions                     3182 non-null   object
 5   Average engagement time per session  3182 non-null   object
 6   Engaged sessions per user            3182 non-null   object
 7   Events per session                   3182 non-null   object
 8   Engagement rate                      3182 non-null   object
 9   Event count                          3182 non-null   object
dtypes: object(10)
memory usage: 248.7+ KB
```

In [27]:
```python
df["DateHour"] = pd.to_datetime(df["DateHour"], format="%Y%m%d%H", errors='coerce')
df.head()
```

Out[27]:

	channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count
0	Direct	2024-04-16 23:00:00	237	300	144	47.526666666666700	0.6075949367088610	4.673333333333330	0.48	1402
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.09737827715360	0.6346153846153850	4.295880149812730	0.4943820224719100	1147
2	Direct	2024-04-17 23:00:00	188	233	115	39.93991416309010	0.6117021276595740	4.587982832618030	0.49356223175965700	1069
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.16015625	0.6684491978609630	4.078125	0.48828125	1044
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552036199100	0.64	4.529411764705880	0.5067873303167420	1001

In [28]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 10 columns):
 #   Column                               Non-Null Count  Dtype         
---  ------                               --------------  -----         
 0   channel group                        3182 non-null   object        
 1   DateHour                             3182 non-null   datetime64[ns]
 2   Users                                3182 non-null   object        
 3   Sessions                             3182 non-null   object        
 4   Engaged sessions                     3182 non-null   object        
 5   Average engagement time per session  3182 non-null   object        
 6   Engaged sessions per user            3182 non-null   object        
 7   Events per session                   3182 non-null   object        
 8   Engagement rate                      3182 non-null   object        
 9   Event count                          3182 non-null   object        
dtypes: datetime64[ns](1), object(9)
memory usage: 248.7+ KB
```

In [29]:
```python
numeric_cols = df.columns.drop(["channel group", "DateHour"])
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric, errors='coerce')
df["Hour"] = df["DateHour"].dt.hour
```

In [30]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3182 entries, 0 to 3181
Data columns (total 11 columns):
 #   Column                               Non-Null Count  Dtype         
---  ------                               --------------  -----         
 0   channel group                        3182 non-null   object        
 1   DateHour                             3182 non-null   datetime64[ns]
 2   Users                                3182 non-null   int64         
 3   Sessions                             3182 non-null   int64         
 4   Engaged sessions                     3182 non-null   int64         
 5   Average engagement time per session  3182 non-null   float64       
 6   Engaged sessions per user            3182 non-null   float64       
 7   Events per session                   3182 non-null   float64       
 8   Engagement rate                      3182 non-null   float64       
 9   Event count                          3182 non-null   int64         
 10  Hour                                 3182 non-null   int32         
dtypes: datetime64[ns](1), float64(4), int32(1), int64(4), object(1)
memory usage: 261.2+ KB
```

In [31]:
```python
df.head()
```

Out[31]:

	channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count	Hour
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402	23
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.295880	0.494382	1147	19
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069	23
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044	18
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001	20

In [32]:
```python
df.describe()
```

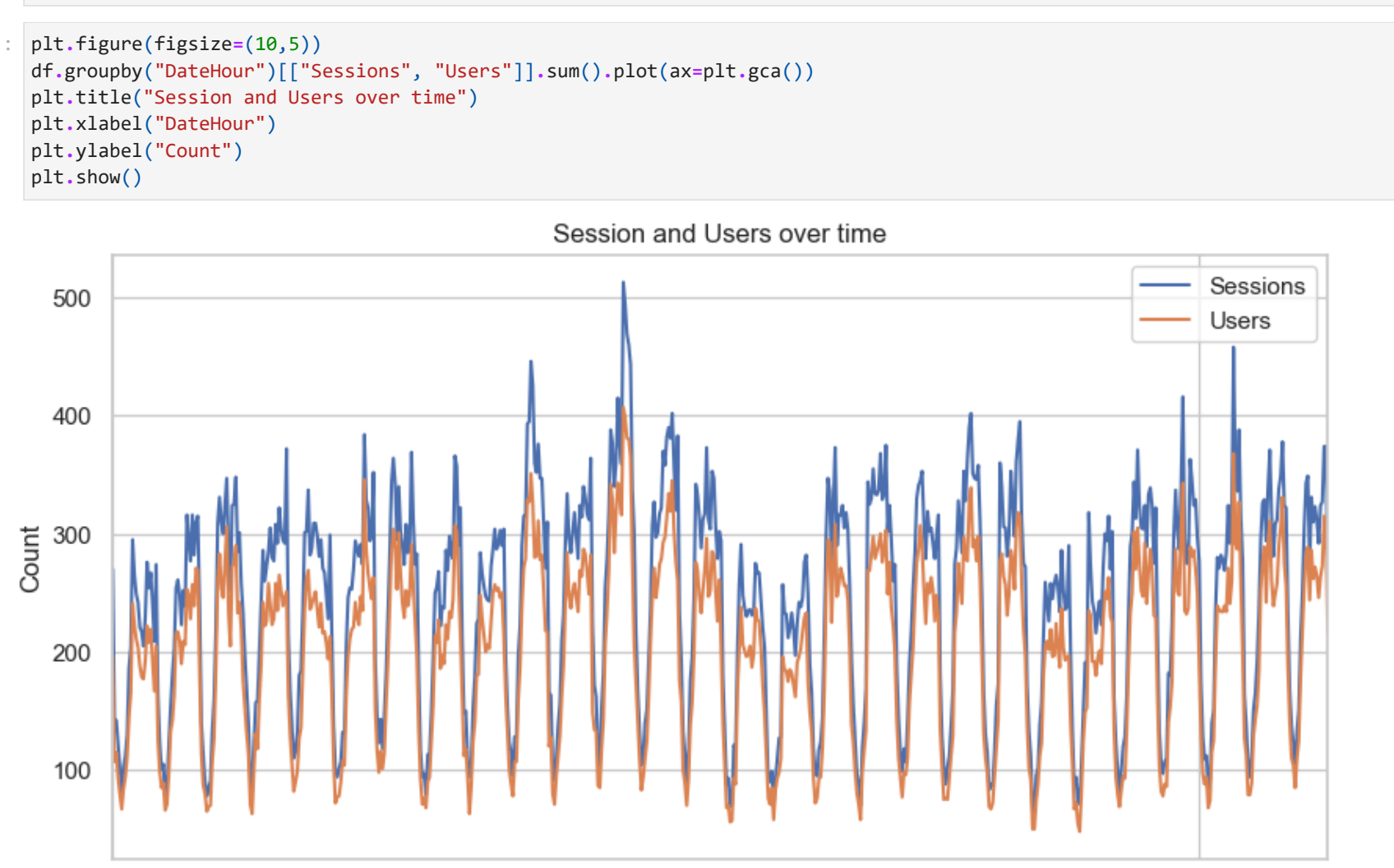
Out[32]:

	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count	Hour
count	3182	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000	3182.000000
mean	2024-04-20 01:17:07.278441216	41.935889	51.192646	28.325581	66.644581	0.606450	4.675969	0.503396	242.272470	11.807040
min	2024-04-06 00:00:00	0.000000	1.000000	0.000000	0.000000	0.000000	1.000000	0.000000	1.000000	0.000000
25%	2024-04-13 02:15:00	20.000000	24.000000	13.000000	32.103034	0.561404	3.750000	0.442902	103.000000	6.000000
50%	2024-04-20 02:00:00	42.000000	51.000000	27.000000	49.020202	0.666667	4.410256	0.545455	226.000000	12.000000
75%	2024-04-26 22:00:00	60.000000	71.000000	41.000000	71.487069	0.750000	5.217690	0.633333	339.000000	18.000000
max	2024-05-03 23:00:00	237.000000	300.000000	144.000000	4525.000000	2.000000	56.000000	1.000000	1402.000000	23.000000
std	NaN	29.582258	36.919962	20.650569	127.200659	0.264023	2.795228	0.228206	184.440313	6.886686

- What patterns or trends can you observe in website sessions and users over time?

In [33]:
```python
sns.set(style="whitegrid")
```

In [34]:
```python
plt.figure(figsize=(10,5))
df.groupby("DateHour")[["Sessions", "Users"]].sum().plot(ax=plt.gca())
plt.title("Session and Users over time")
plt.xlabel("DateHour")
plt.ylabel("Count")
plt.show()
```



- Total users by Channel

In [35]:
```python
df.columns
```

Out[35]:
```
Index(['channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',
       'Average engagement time per session', 'Engaged sessions per user',
       'Events per session\t', 'Engagement rate', 'Event count', 'Hour'],
      dtype='object')
```

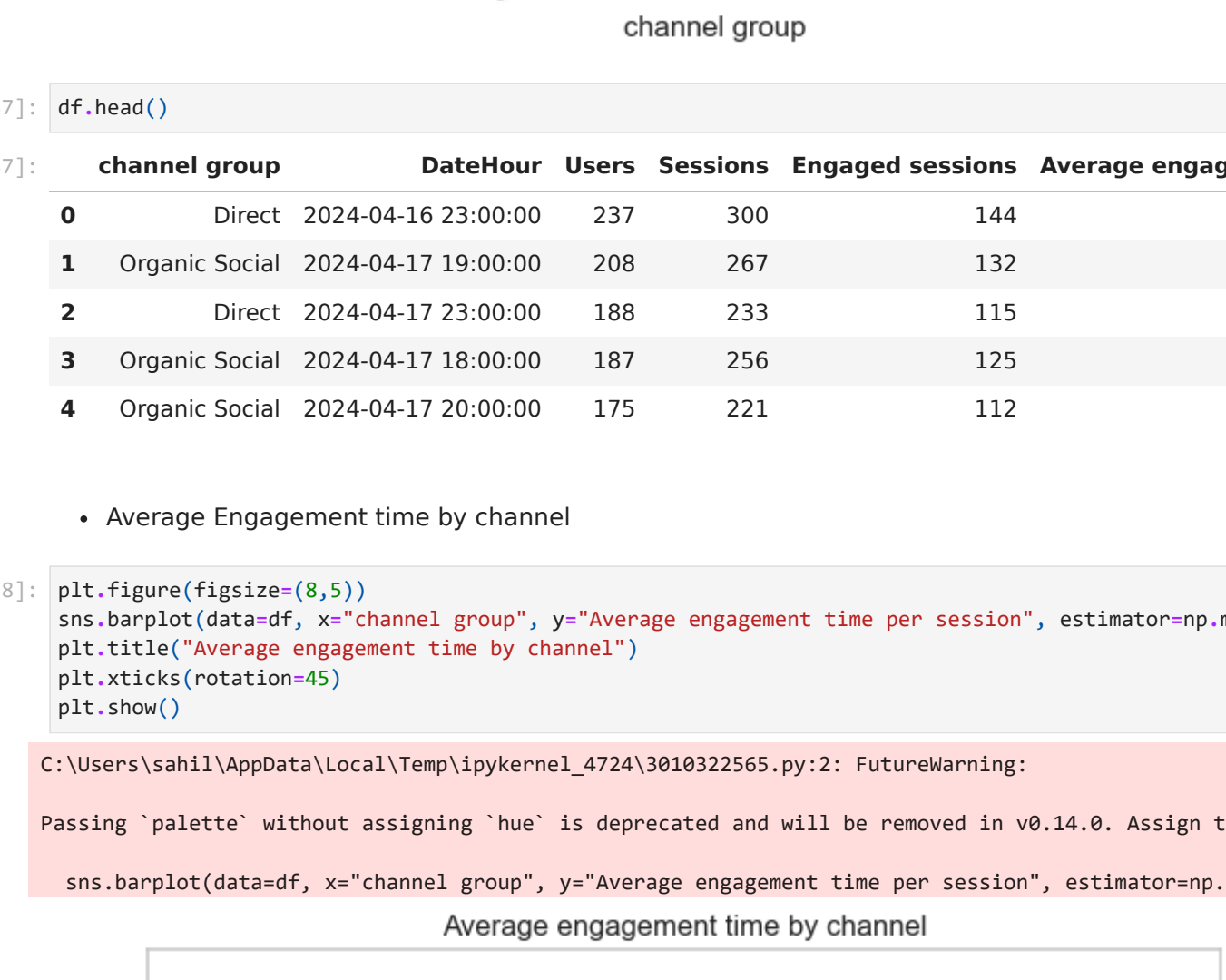
In [36]:
```python
import seaborn as sns

plt.figure(figsize=(8,5))
sns.barplot(data=df, x="channel group", y="Users", estimator=np.sum, palette="viridis")
plt.title("Total Users by channel")
plt.xticks(rotation=45)
plt.show()
```

```
C:\Users\sahil\AppData\Local\Temp\ipykernel_4724\2376754780.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=df, x="channel group", y="Users", estimator=np.sum, palette="viridis")
```



In [37]:
```python
df.head()
```

Out[37]:

	channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count	Hour
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402	23
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.295880	0.494382	1147	19
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069	23
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044	18
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001	20

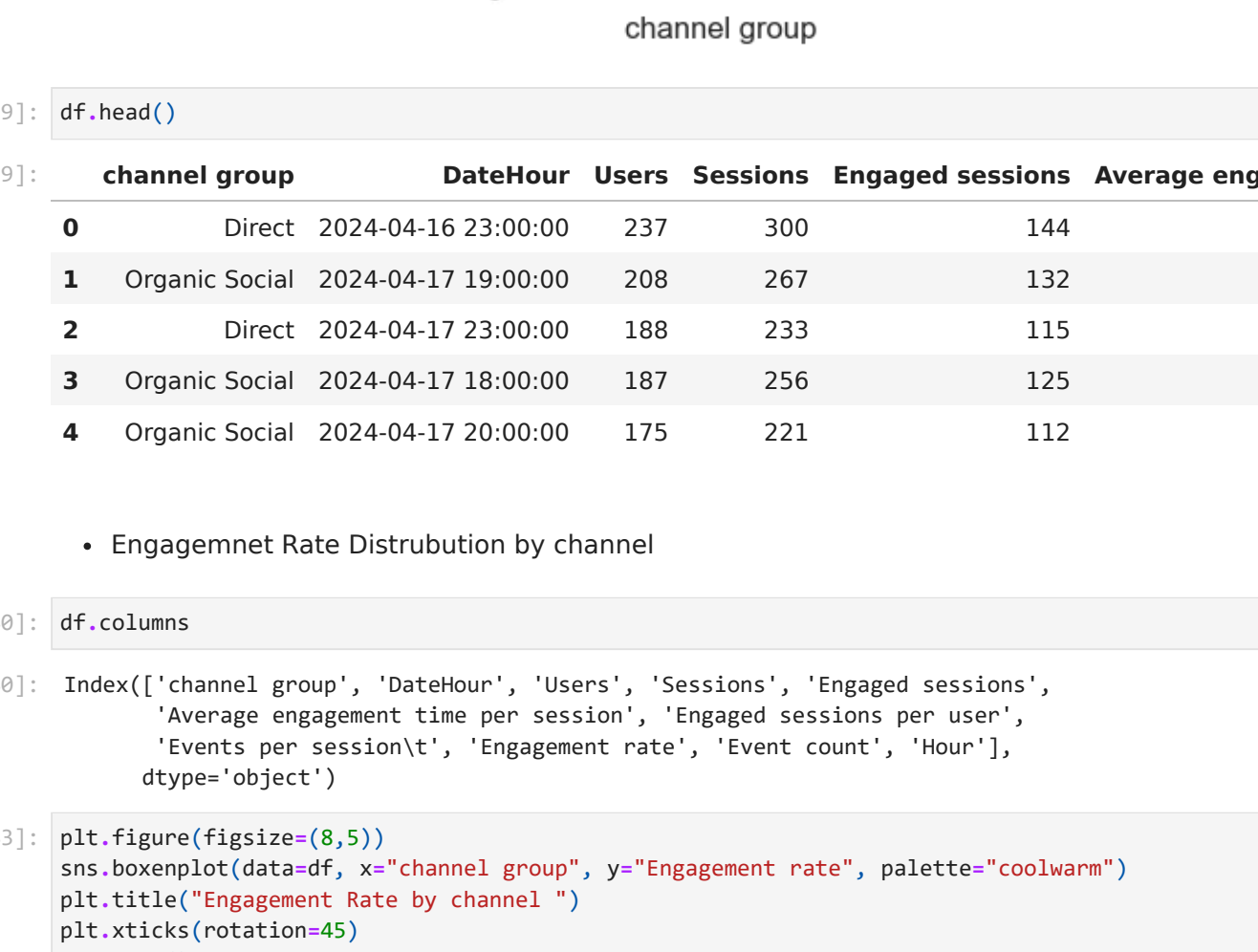
- Average Engagement time by channel

In [38]:
```python
plt.figure(figsize=(8,5))
sns.barplot(data=df, x="channel group", y="Average engagement time per session", estimator=np.mean, palette="magma")
plt.title("Average engagement time by channel")
plt.xticks(rotation=45)
plt.show()
```

```
C:\Users\sahil\AppData\Local\Temp\ipykernel_4724\3010322565.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(data=df, x="channel group", y="Average engagement time per session", estimator=np.mean, palette="magma")
```



In [39]:
```python
df.head()
```

Out[39]:

	channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count	Hour
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402	23
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.295880	0.494382	1147	19
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069	23
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044	18
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001	20

- Engagemnet Rate Distrubution by channel

In [40]:
```python
df.columns
```

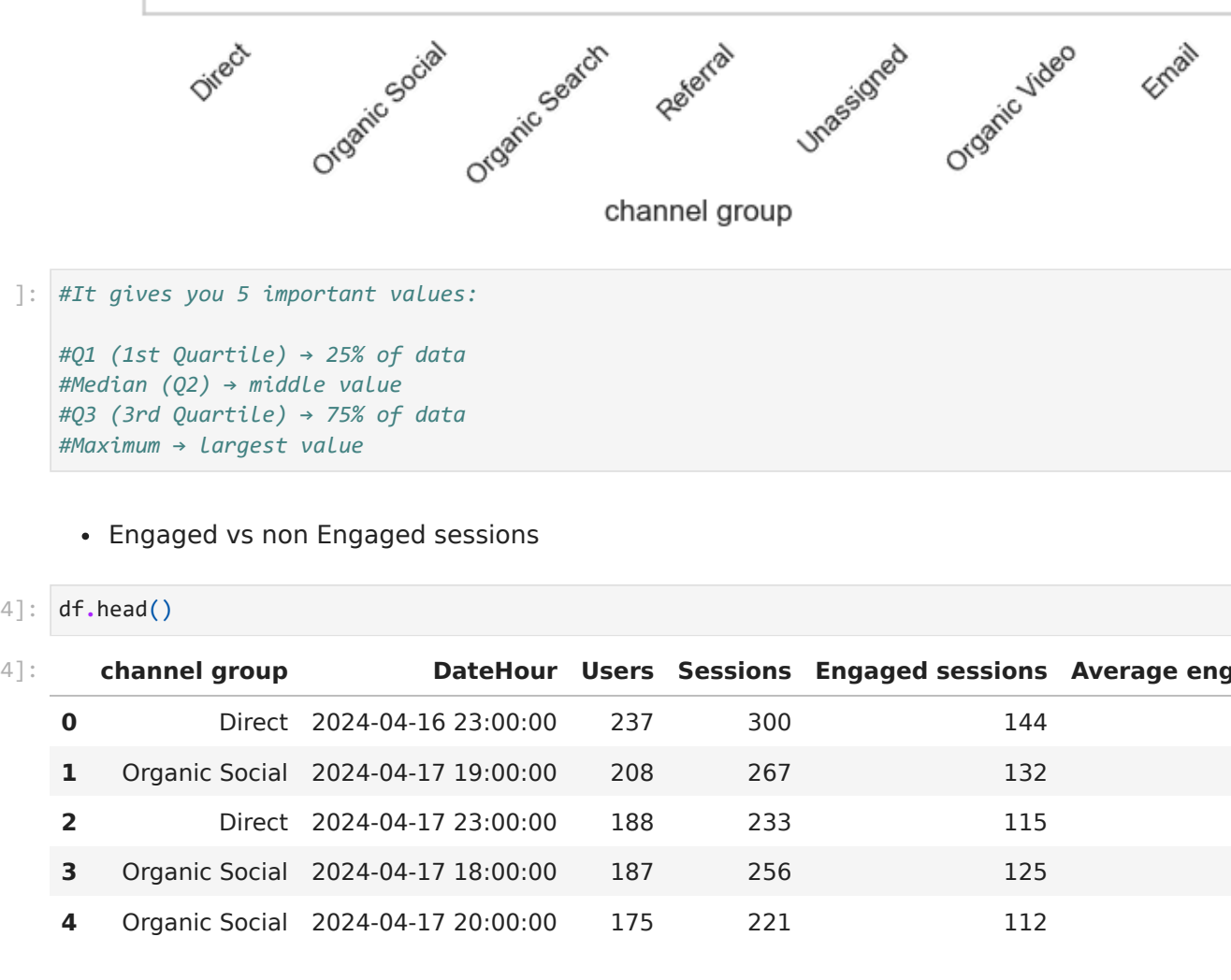
Out[40]:
```
Index(['channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',
       'Average engagement time per session', 'Engaged sessions per user',
       'Events per session\t', 'Engagement rate', 'Event count', 'Hour'],
      dtype='object')
```

In [43]:
```python
plt.figure(figsize=(8,5))
sns.boxenplot(data=df, x="channel group", y="Engagement rate", palette="coolwarm")
plt.title("Engagement Rate by channel ")
plt.xticks(rotation=45)
plt.show()
```

```
C:\Users\sahil\AppData\Local\Temp\ipykernel_4724\3771661893.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.boxenplot(data=df, x="channel group", y="Engagement rate", palette="coolwarm")
```



In [ ]:
```python
#It gives you 5 important values:

#Q1 (1st Quartile) → 25% of data
#Median (Q2) → middle value
#Q3 (3rd Quartile) → 75% of data
#Maximum → Largest value
```

- Engaged vs non Engaged sessions

In [44]:
```python
df.head()
```

Out[44]:

	channel group	DateHour	Users	Sessions	Engaged sessions	Average engagement time per session	Engaged sessions per user	Events per session\t	Engagement rate	Event count	Hour
0	Direct	2024-04-16 23:00:00	237	300	144	47.526667	0.607595	4.673333	0.480000	1402	23
1	Organic Social	2024-04-17 19:00:00	208	267	132	32.097378	0.634615	4.295880	0.494382	1147	19
2	Direct	2024-04-17 23:00:00	188	233	115	39.939914	0.611702	4.587983	0.493562	1069	23
3	Organic Social	2024-04-17 18:00:00	187	256	125	32.160156	0.668449	4.078125	0.488281	1044	18
4	Organic Social	2024-04-17 20:00:00	175	221	112	46.918552	0.640000	4.529412	0.506787	1001	20

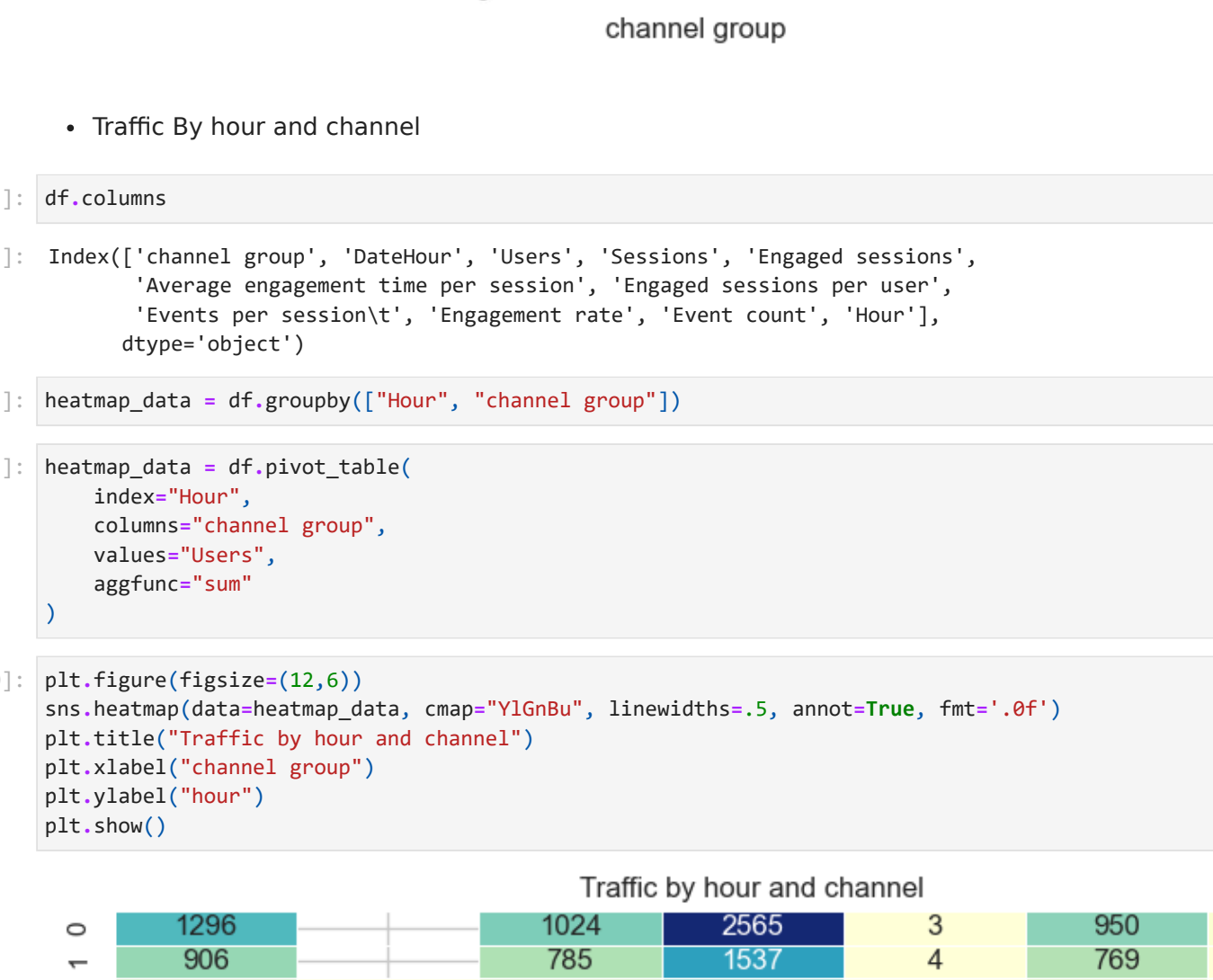
In [45]:
```python
df.columns
```

Out[45]:
```
Index(['channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',
       'Average engagement time per session', 'Engaged sessions per user',
       'Events per session\t', 'Engagement rate', 'Event count', 'Hour'],
      dtype='object')
```

In [49]:
```python
session_df = df.groupby("channel group")[["Sessions", "Engaged sessions"]].sum().reset_index()
session_df["Non-Engaged"] = session_df["Sessions"] - session_df["Engaged sessions"]
session_df_melted = session_df.melt(id_vars="channel group", value_vars=["Engaged sessions", "Non-Engaged"])

#df.melt() is a Pandas function used to reshape data from wide format → long format.
```

In [50]:
```python
plt.figure(figsize=(8,5))
sns.barplot(data=session_df_melted, x="channel group", y="value", hue="variable")
plt.title("Engaged vs Non Engaged Session")
plt.xticks(rotation=45)
plt.show()
```



- Traffic By hour and channel

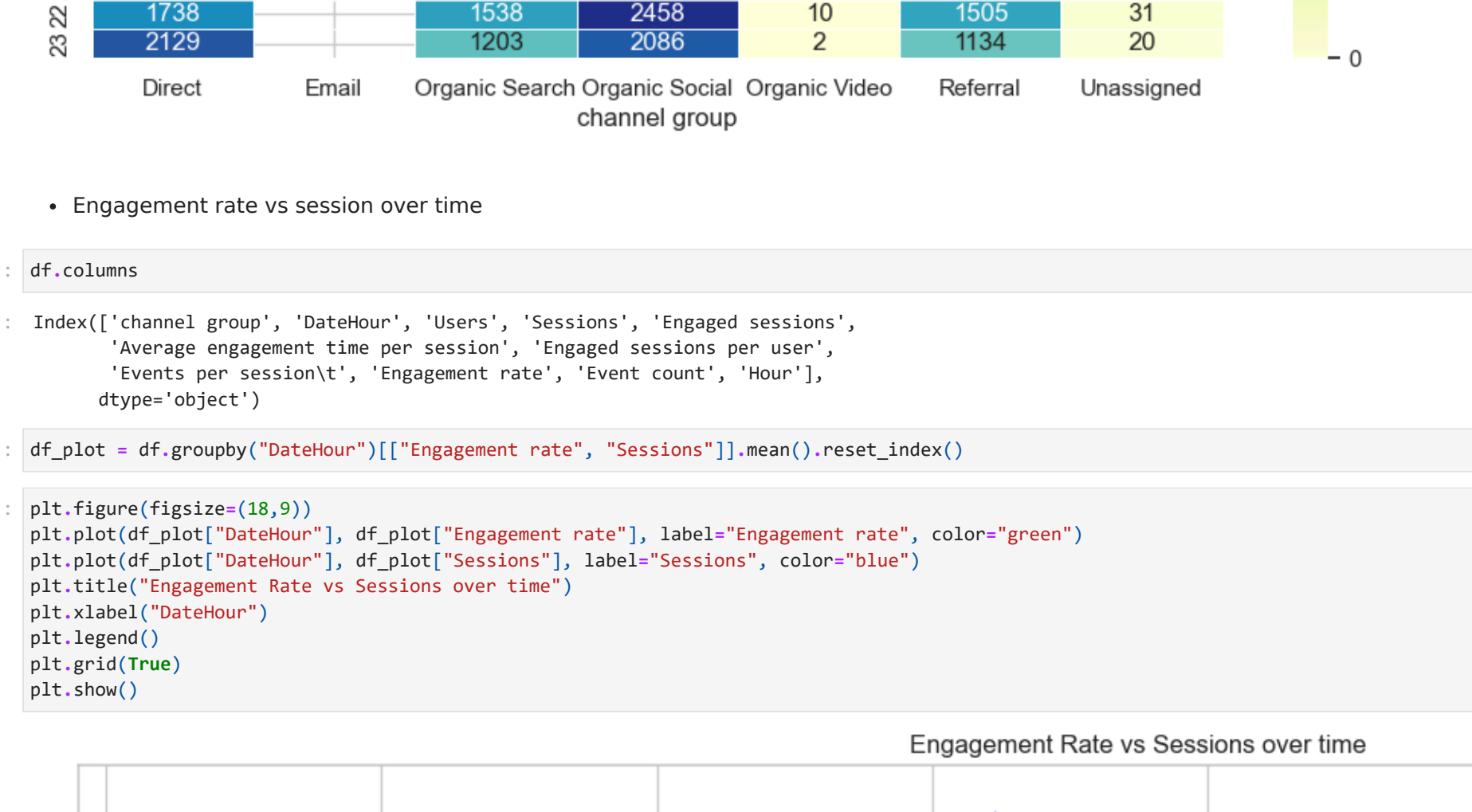
In [51]:
```python
df.columns
```

Out[51]:
```
Index(['channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',
       'Average engagement time per session', 'Engaged sessions per user',
       'Events per session\t', 'Engagement rate', 'Event count', 'Hour'],
      dtype='object')
```

In [53]:
```python
heatmap_data = df.groupby(["Hour", "channel group"])
```

In [59]:
```python
heatmap_data = df.pivot_table(
    index="Hour",
    columns="channel group",
    values="Users",
    aggfunc="sum"
)
```

In [60]:
```python
plt.figure(figsize=(12,6))
sns.heatmap(data=heatmap_data, cmap="YlGnBu", linewidths=.5, annot=True, fmt='.0f')
plt.title("Traffic by hour and channel")
plt.xlabel("channel group")
plt.ylabel("hour")
plt.show()
```

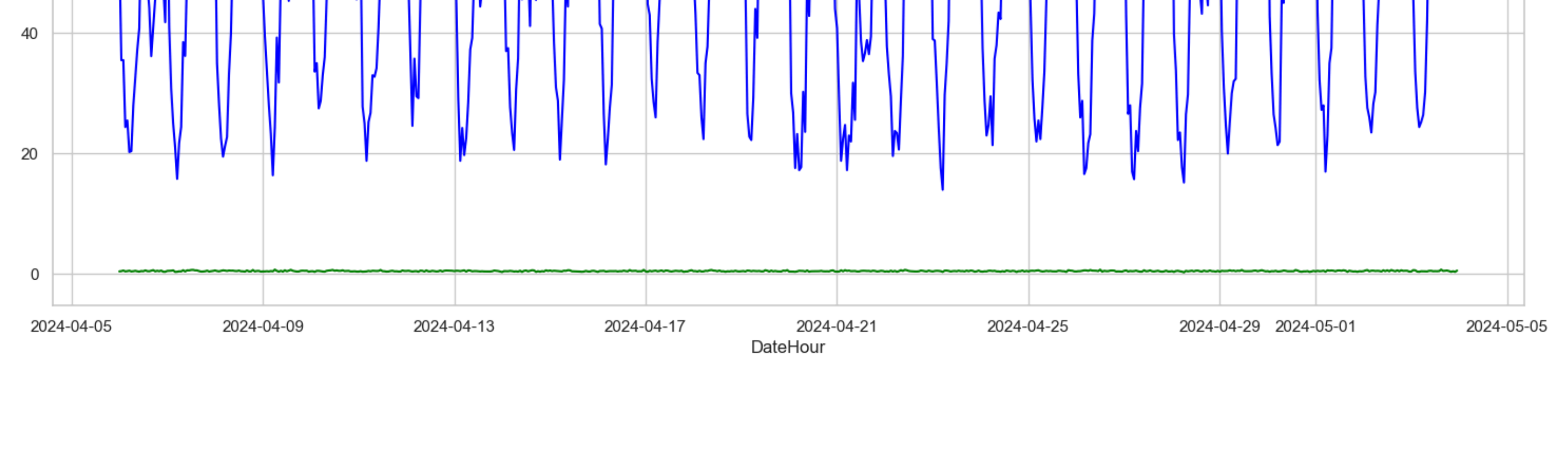


- Engagement rate vs session over time

In [63]:
```python
df.columns
```

Out[63]:
```
Index(['channel group', 'DateHour', 'Users', 'Sessions', 'Engaged sessions',
       'Average engagement time per session', 'Engaged sessions per user',
       'Events per session\t', 'Engagement rate', 'Event count', 'Hour'],
      dtype='object')
```

In [64]:
```python
df_plot = df.groupby("DateHour")[["Engagement rate", "Sessions"]].mean().reset_index()
```

In [69]:
```python
plt.figure(figsize=(18,9))
plt.plot(df_plot["DateHour"], df_plot["Engagement rate"], label="Engagement rate", color="green")
plt.plot(df_plot["DateHour"], df_plot["Sessions"], label="Sessions", color="blue")
plt.title("Engagement Rate vs Sessions over time")
plt.xlabel("DateHour")
plt.legend()
plt.grid(True)
plt.show()
```